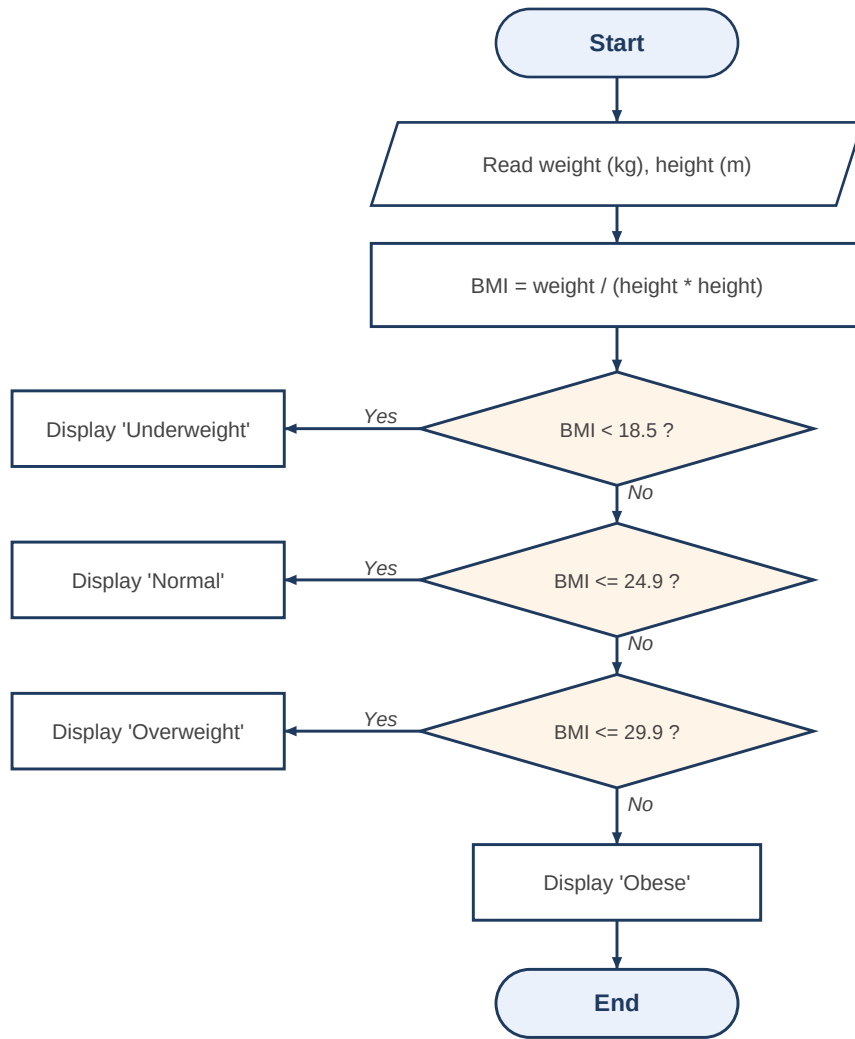


Flowgorithm Flowchart & Pseudocode Assignments

1. BMI Calculator and Classification
2. Checking if an Array is Sorted in Ascending Order
3. Food-Ordering System with Regular / Premium Delivery Charges
(including class inheritance relationship)

Assignment 1: BMI Calculator

Flowchart — read weight & height, compute BMI, classify result



Assignment 1: BMI Calculator

Pseudocode

Pseudocode

```
Program BMI_Calculator
  Declare Real weight, height, bmi

  Display "Enter weight in kilograms: "
  Input weight
  Display "Enter height in metres: "
  Input height

  Set bmi = weight / (height * height)

  If bmi < 18.5 Then
    Display "Category: Underweight"
  Else If bmi <= 24.9 Then
    Display "Category: Normal"
  Else If bmi <= 29.9 Then
    Display "Category: Overweight"
  Else
    Display "Category: Obese"
  End If

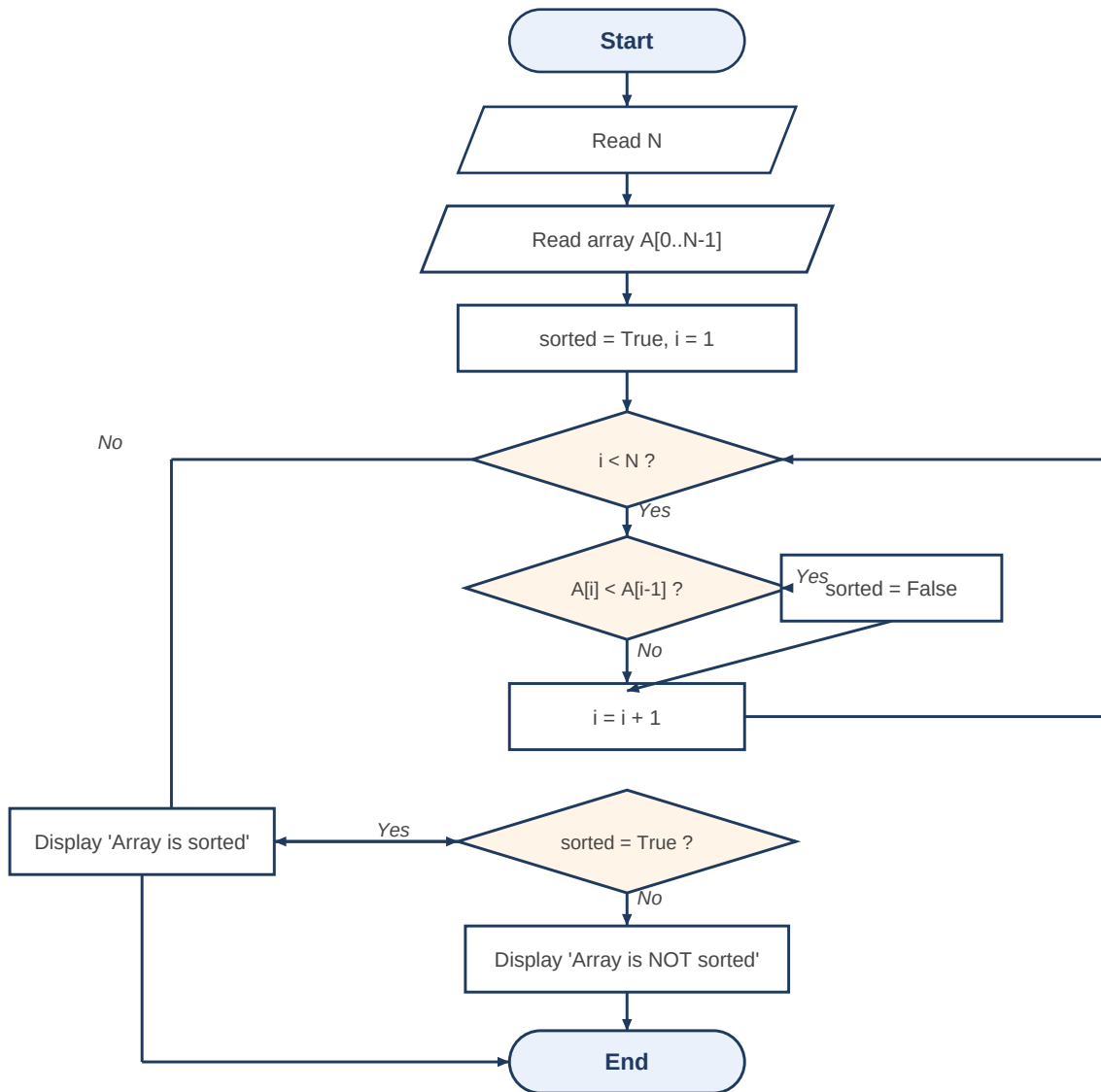
  Display "Your BMI is: ", bmi
End Program
```

Notes

- BMI = weight (kg) / height (m)² — height must be in metres, not centimetres.
- The four ranges are checked in order using Else If, so once a condition is True the rest are skipped.
- Standard WHO cut-offs used here: <18.5 Underweight, 18.5-24.9 Normal, 25-29.9 Overweight, >=30 Obese.

Assignment 2: Array Sorted Check

Flowchart — read N values, check ascending order



Assignment 2: Array Sorted Check

Pseudocode

Pseudocode

```
Program Check_Sorted_Array
  Declare Integer N, i
  Declare Real A[100]
  Declare Boolean sorted

  Display "Enter number of elements N: "
  Input N

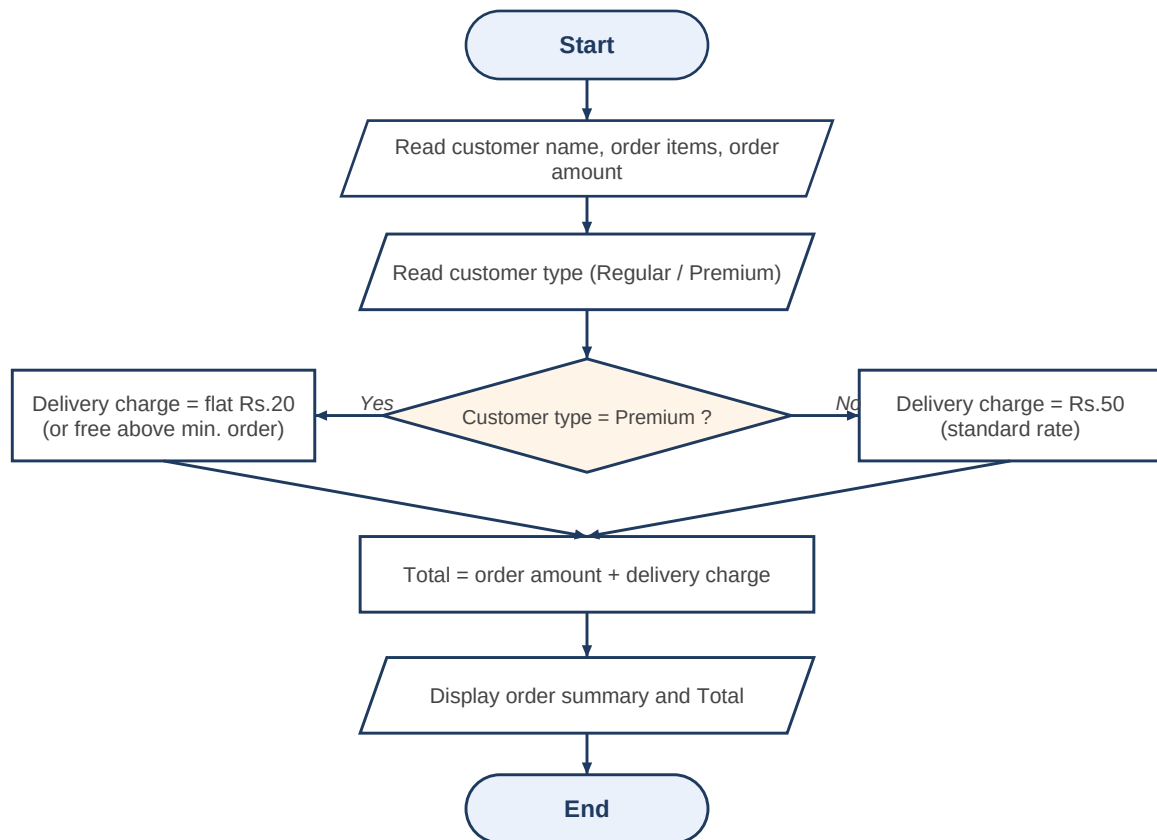
  For i = 0 To N - 1
    Display "Enter A[, i, "]: "
    Input A[i]
  End For

  Set sorted = True
  For i = 1 To N - 1
    If A[i] < A[i - 1] Then
      Set sorted = False
    End If
  End For

  If sorted = True Then
    Display "Array is sorted in ascending order"
  Else
    Display "Array is NOT sorted in ascending order"
  End If
End Program
```

Assignment 3: Food-Ordering System

Flowchart — order details + delivery charge by customer type



Assignment 3: Food-Ordering System

Pseudocode

Pseudocode

```
Program Food_Order
  Declare String customerName, customerType
  Declare Real orderAmount, deliveryCharge, total

  Display "Enter customer name: "
  Input customerName
  Display "Enter order amount: "
  Input orderAmount
  Display "Enter customer type (Regular / Premium): "
  Input customerType

  If customerType = "Premium" Then
    Set deliveryCharge = 20
  Else
    Set deliveryCharge = 50
  End If

  Set total = orderAmount + deliveryCharge

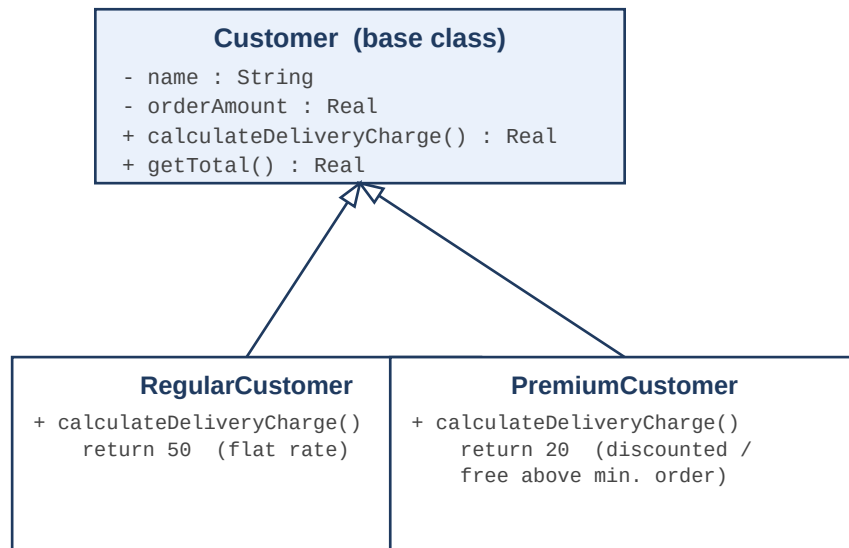
  Display "Customer: ", customerName
  Display "Order Amount: ", orderAmount
  Display "Delivery Charge: ", deliveryCharge
  Display "Total Payable: ", total
End Program
```

Assignment 3: Inheritance Relationship

How Regular and Premium customers relate as classes

Why inheritance fits this problem

Regular and Premium customers share the same core order details (name, items, order amount) but differ only in how the delivery charge is calculated. This is a classic case for inheritance: put the shared data/behaviour in one base class, and let each customer type override just the part that differs (the delivery-charge rule).



Hollow-triangle arrows point from each subclass up to the base class it inherits from (UML 'is-a' relationship).

Assignment 3: Inheritance Relationship

Pseudocode (class-based)

Pseudocode

```
Class Customer
  Declare String name
  Declare Real orderAmount

  Method calculateDeliveryCharge() : Real
    // overridden by each subclass
    Return 0
  End Method

  Method getTotal() : Real
    Return orderAmount + calculateDeliveryCharge()
  End Method
End Class

Class RegularCustomer Inherits Customer
  Method calculateDeliveryCharge() : Real
    Return 50          // standard flat rate
  End Method
End Class

Class PremiumCustomer Inherits Customer
  Method calculateDeliveryCharge() : Real
    If orderAmount >= 500 Then
      Return 0          // free delivery above minimum
    Else
      Return 20         // discounted rate
    End If
  End Method
End Class

Program Main
  Declare Customer c
  Display "Customer type (Regular/Premium): "
  Input custType
  If custType = "Premium" Then
    Set c = New PremiumCustomer()
  Else
    Set c = New RegularCustomer()
  End If
  Input c.name, c.orderAmount
  Display "Total Payable: ", c.getTotal()
End Program
```